

A c_0 -VECTOR OUTSIDE THE CLOSED SPAN IN A VARIABLE-EXPONENT SEQUENTIAL SPACE

AGENT LANE 05

SOURCE AND TARGET

The source paper is Talponen, *Constructions of sequential spaces*, arXiv:0905.0812 [1]. It introduces the sequential variable-exponent spaces

$$\ell^{p(\cdot)} = \{x \in \ell^\infty : \Phi(x) < \infty\},$$

where, for $p : \mathbb{N} \rightarrow [1, \infty]$,

$$|||x|||(1) = |x_1| \boxplus_{p(1)} |x_2|, \quad |||x|||(k) = |||x|||(k-1) \boxplus_{p(k)} |x_{k+1}|,$$

and $\Phi(x) = \lim_k |||x|||(k)$. Here $a \boxplus_r b = (a^r + b^r)^{1/r}$ for finite r .

On page 8 the paper repeats the following open problem:

Does $\ell^{p(\cdot)} \cap c_0$ always coincide with $[(e_n)] \subset \ell^{p(\cdot)}$?

$$\lim_{n \rightarrow \infty} d_{\text{BM}}(Q_n(\ell^{p(\cdot)}), \ell^1) = 1.$$

Finally we reiterate the open problems that have arisen in this note:

- We do not know if the space in Example 3.1 is MLUR.
- Does $\ell^{p(\cdot)} \cap c_0$ always coincide with $[(e_n)] \subset \ell^{p(\cdot)}$?
- Given $p : \mathbb{N} \rightarrow (0, \infty)$, is $\ell^{p(\cdot)}$ necessarily strictly convex or smooth?

RESULT

The answer is negative. We construct a finite-valued exponent map $p : \mathbb{N} \rightarrow [2, \infty)$ and a vector $x \in \ell^{p(\cdot)} \cap c_0$ such that $x \notin [(e_n)]$.

PROOF INTUITION

The obstruction is that membership in the closed span $[(e_n)]$ requires the coordinate tails $Q_n x = x - P_n x$ to vanish in the $\ell^{p(\cdot)}$ norm. We build x from blocks. Inside block j the exponent is very large, $r_j = j + 1$, and there are j^{r_j} equal coordinates c/j . Viewed by itself, the block has norm exactly c , so tails starting at the block never become small. Viewed after the previously accumulated initial segment, however, the same block contributes only c^{r_j} under an r_j -root. Since $\sum_j c^{r_j}/r_j < \infty$, the global recursive norm remains bounded.

Lemma 1. *For every $p : \mathbb{N} \rightarrow [1, \infty]$ and every $y \in \ell^{p(\cdot)}$,*

$$y \in [(e_n)] \iff \|Q_n y\|_{\ell^{p(\cdot)}} \rightarrow 0.$$

Date: June 15, 2026.

Proof. The implication from right to left is immediate because $P_n y$ is finitely supported and

$$y - P_n y = Q_n y \rightarrow 0.$$

Conversely, suppose $y \in [(e_n)]$. Given $\varepsilon > 0$, choose a finitely supported vector z , supported in $\{1, \dots, N\}$, such that $\|y - z\|_{\ell^{p(\cdot)}} < \varepsilon$. For $n \geq N$, $Q_n z = 0$. The recursive norm is coordinatewise monotone in the absolute values of the coordinates, so every coordinate projection, including Q_n , is contractive. Hence

$$\|Q_n y\|_{\ell^{p(\cdot)}} = \|Q_n(y - z)\|_{\ell^{p(\cdot)}} \leq \|y - z\|_{\ell^{p(\cdot)}} < \varepsilon$$

for all $n \geq N$. □

Theorem 2. *There are $p : \mathbb{N} \rightarrow [2, \infty)$ and*

$$x \in \ell^{p(\cdot)} \cap c_0$$

such that $x \notin [(e_n)]$. Therefore $\ell^{p(\cdot)} \cap c_0$ need not coincide with $[(e_n)] \subset \ell^{p(\cdot)}$.

Proof. Fix $0 < c < 1$, for instance $c = 1/2$. Put

$$r_j = j + 1, \quad m_j = j^{r_j} \quad (j \geq 1).$$

Let $N_1 = 2$ and $N_{j+1} = N_j + m_j$. The j -th block of coordinates is

$$B_j = \{N_j, N_j + 1, \dots, N_{j+1} - 1\}.$$

Define $x \in \ell^\infty$ by

$$x_1 = 1, \quad x_n = \frac{c}{j} \quad \text{if } n \in B_j.$$

Since $c/j \rightarrow 0$, $x \in c_0$.

Define $p : \mathbb{N} \rightarrow [2, \infty)$ by

$$p(k) = r_j \quad \text{whenever } N_j - 1 \leq k \leq N_{j+1} - 2.$$

These intervals partition $\mathbb{N}$, so p is well defined.

Let A_j be the norm of the initial segment consisting of x_1 and the first j whole blocks; equivalently

$$A_j = \|P_{N_{j+1}-1} x\|_{\ell^{p(\cdot)}}.$$

Then $A_0 = 1$. Since every exponent used while appending block j is r_j ,

$$A_j = \left(A_{j-1}^{r_j} + m_j \left(\frac{c}{j} \right)^{r_j} \right)^{1/r_j} = \left(A_{j-1}^{r_j} + c^{r_j} \right)^{1/r_j}.$$

As $A_{j-1} \geq 1$,

$$A_j = A_{j-1} \left(1 + \left(\frac{c}{A_{j-1}} \right)^{r_j} \right)^{1/r_j} \leq A_{j-1} (1 + c^{r_j})^{1/r_j}.$$

Therefore

$$A_j \leq \prod_{i=1}^j (1 + c^{r_i})^{1/r_i}.$$

The infinite product converges because

$$\sum_{i=1}^{\infty} \frac{\log(1 + c^{r_i})}{r_i} \leq \sum_{i=1}^{\infty} \frac{c^{r_i}}{r_i} < \infty.$$

All finite prefixes inside a block have norm at most the norm after the whole block. Hence $\sup_n \|P_n x\|_{\ell^{p(\cdot)}} < \infty$, so $x \in \ell^{p(\cdot)}$.

It remains to show that $x \notin [(e_n)]$. Consider the tail starting just before block j , namely $Q_{N_j-1}x$. The finite projection of this tail onto the coordinates in B_j has all earlier coordinates equal to zero, then m_j coordinates equal to c/j , and all relevant exponents equal to r_j . Its norm is

$$\left(m_j \left(\frac{c}{j} \right)^{r_j} \right)^{1/r_j} = c.$$

By coordinatewise monotonicity,

$$\|Q_{N_j-1}x\|_{\ell^{p(\cdot)}} \geq c$$

for every j . Thus the tails of x do not tend to zero in norm. Lemma 1 gives $x \notin [(e_n)]$. $\square$

VERIFICATION NOTES

The construction uses only finite exponents $p(k) \geq 2$, so it fits the source paper’s setting $p : \mathbb{N} \rightarrow [1, \infty]$, and also the broader $(0, \infty)$ -valued wording of the final open-problem list.

A small script, `code/block_counterexample_check.py`, prints the first few values of the block recurrence

$$A_j = (A_{j-1}^{j+1} + c^{j+1})^{1/(j+1)}$$

and the fixed lower bound c for the selected block tails. This is only a sanity check for the displayed formulas; the convergence proof above is analytic.

NOVELTY AND SEARCH BOUNDS

Before promotion, the run’s registry, solution, attempt, and proof-gap indexes were searched for arXiv:0905.0812, the paper title, Talponen, and the core phrases around $\ell^{p(\cdot)} \cap c_0$ and $[(e_n)]$. No existing solution packet or serious prior attempt for this question was found.

External web searches on 2026-06-15 used exact and near-exact phrases including

“Constructions of sequential spaces” Talponen “c_0” “[(e_n)]”,
“ell^p(\cdot)” “c_0” “[(e_n)]”,

and

“variable exponent sequential spaces” Talponen.

These searches surfaced the source paper but no later explicit answer to this equality question. The packet should therefore be read as a candidate new

counterexample with medium novelty confidence, pending broader human literature review.

HUMAN REVIEW RECOMMENDATION

Review as a candidate full negative answer to the second open problem in [1]. The main checks are the indexing of the block exponents, the contractivity of coordinate projections under the recursive norm, and the tail-norm characterization in Lemma 1.

REFERENCES

- [1] J. Talponen, *Constructions of sequential spaces*, arXiv:0905.0812, 2009.